Sparse Traffic Accident Risk Forecasting With Spatial-Temporal Knowledge Graphs

Shengnan Guo, Yan Lin, Wei Chen, Weiwen Tang, Haochen Lv, Rongzhi Zhou, Junliang Lin, Youfang Lin, and Huaiyu Wan

***Abstract*—Accurate prediction of traffic accident risk at both the road segment and taxi zone granularities is essential for modern intelligent transportation systems. Such predictions offer early warnings to travelers and transportation authorities, contributing to safer mobility. However, achieving accurate traffic accident risk prediction faces two major challenges. Firstly, traffic accident risk is affected by multi-source, entangled and dynamic factors. Accurate accident risk prediction requires a deep understanding of the interactions among these factors, while existing methods struggle to address this task effectively. Secondly, the sparsity of traffic accident data leads to the zero-inflation issue, causing models to appear effective on risk regression metrics while failing to accurately identify high-risk regions. To address these challenges, we propose STKGRisk, a novel model that simultaneously predicts traffic accident risk for both road segments and taxi zones. Specifically, we construct the first spatial-temporal knowledge graph for traffic accident risk analysis and design a diachronic embedding module to capture the high-order, dynamic interactions between multi-source factors and traffic accidents. Building upon these embeddings, we develop a spatial-temporal graph network encoder to capture the spatial-temporal correlations of accident risk for two granularities from multi-level and multi-view perspectives. To tackle the zero-inflation issue, we design a Zero-Inflated Mixture Poisson decoder to learn the occurrence patterns of sparse accident risk data. Extensive experiments on three real-world traffic accident datasets demonstrate that STKGRisk outperforms existing models. STKGRisk almost achieves the best predictive performance on both the segment- and zone-granularity accident risk prediction tasks, and excels particularly in the identification of high-risk areas.**

***Index Terms*—Spatial-temporal data mining, traffic accident risk prediction, spatial-temporal knowledge graph, zero-inflation mixture poisson.**

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I. Introduction

Accurate traffic accident risk forecasting is essential for modern intelligent transportation systems. In real-world applications, particular attention is paid to accident risks at two spatial granularities, including road segments and taxi zones, due to their practical significance for both individual travelers and transportation authorities. These two granularities of traffic accident risk can provide early-warning signals to mitigate traffic accident impact. For example, when the accident risk of some road segments is predicted to surge in the next hour, the administrator could broadcast safe driving guidelines in advance to the travelers passing on these road segments, and send a warning message to the travelers planning to pass on these road segments. Similarly, when a taxi zone is predicted to have a surge in accident risk, the administrator can prevent serious traffic accidents in advance by dispatching more policemen to patrol and strengthening traffic control. Therefore, accurately forecasting accident risk at both segment and zone levels is vital for minimizing casualties and economic losses caused by traffic accidents. Besides, segment-granularity and zone-granularity accident risk usually possess strong correlations stemming from their spatial associations and common causal factors. Modeling these two granularities jointly can exploit these correlations and enhance the predictive performance for both. However, accurate traffic accident risk prediction remains a challenging task, primarily due to the following reasons.

1) *Traffic accident risk is affected by multi-source, entangled, and dynamic factors:* Accurate accident risk prediction requires understanding the complex interactions among a wide range of factors. Specifically, accident risk is influenced by both current traffic state and historical traffic accident occurrence patterns. Besides, external environmental factors such as weather and surrounding points of interest (POIs) also impact accident risk. These contributing factors are inherently dynamic over time. For example, traffic state and weather vary throughout the day, and their influence on accident risk changes accordingly. Moreover, as shown in Fig. 2, there are not only explicit and one-order intersections (i.e., between weather and taxi zones) but also implicit high-order interactions (i.e., among weather, taxi zones, segments, and accidents) among these factors. Thus, incorporating the dynamic, high-order interactions among these multi-source factors is important for achieving accurate accident risk prediction.

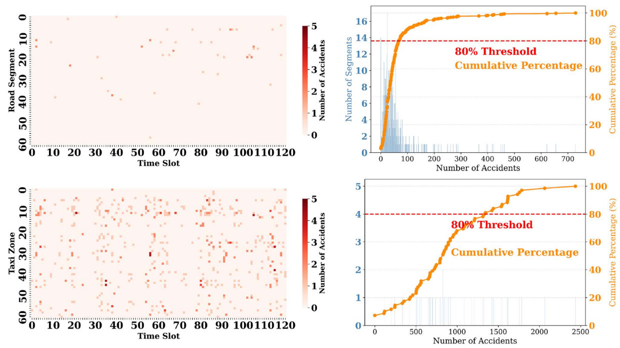


Fig. 1. The first column displays the number of accidents across 60 selected segments and 60 regions for a single day. The second column presents histograms showing the distribution of segments and zones according to their accident frequencies over a year.

To incorporate diverse sources of factors, early studies extend beyond traffic accident data. For instance, SDAE [1] includes human mobility data from GPS traces, while SDCAE [2] incorporates traffic flow data. More recent spatial-temporal graph convolutional models, such as RiskOracle [3], GSNet [4], and RiskSeq [5], integrate weather, temperature, and traffic state information. DF-TAR [6] and MG-TAR [7] go further by leveraging dangerous driving behaviors to enhance predictive power. MGHSTN [8] innovates by incorporating remote sensing data to reveal the backgrounds of regions where traffic accidents happen. While these studies have demonstrated the importance of multi-source factors, most of them adopt simple horizontal concatenation of input features to incorporate external factors. This strategy overlooks the complex interactions among various factors, which are critical for accurate accident risk prediction. This calls for the development of novel modeling techniques that move beyond shallow input feature integration, explicitly modeling the dynamic, high-order interactions among diverse factors to better incorporate their influence on accident risk.

2) *Traffic accident risk prediction seriously suffers from the zero-inflation issue.* As we can see in Fig. 1, traffic accidents are rare events, resulting in a large number of zero values in accident risk data. This will cause model predictions to converge towards zero, also known as the zero-inflation issue. Predictions that converge to zero are of little value for practical applications. Therefore, addressing the zero-inflation problem and obtaining predictions with practical significance is an important challenge that requires further exploration and solution. To mitigate this issue, GSNet proposes a weighted loss function, MVMT-STN [9] adopts a multi-task learning framework to jointly predict multi-granularity traffic accident risks by considering the spatial associations across granularities. MG-TAR leverages dangerous driving data to enhance prediction accuracy. RiskOracle and RiskSeq employ data augmentation techniques to convert zero values into negative values. RiskContra [10] designs a novel contrastive learning approach for augmenting traffic risk samples. However, all of these models rely solely on Mean Squared Error (MSE) as the loss function, which implicitly assumes that the underlying distribution of accident risk data follows a Gaussian distribution. In reality, the underlying distribution of accident risk data deviates from Gaussian due to the abundance of many zero values. As a result, these models fail to solve the zero-inflation issue fundamentally. It is vital to choose a suitable distribution to model and fit the underlying distribution of sparse accident risk data effectively.

To address the above challenges, we propose a novel model called STKGRisk that simultaneously predicts traffic accident risks at both the road segment and taxi zone granularities. Specifically, inspired by the ability of Knowledge Graphs (KGs) to represent high-order interactions between entities [11], and the capability of Temporal Knowledge Graphs (TKGs) to model data dynamics, we first construct a Traffic Accident Spatial Temporal Knowledge Graph (STKG). Compared with conventional approaches that rely on simple feature concatenation or static graph structures, our STKG enables a unified representation of multi-source factors and their evolving relationships, which is crucial for accurately capturing the underlying mechanisms of traffic accident risk. We then design a diachronic embedding module that leverages minute-level temporal data to capture the high-order interactions and dynamics among multi-source factors. Next, we simultaneously encode the spatial-temporal correlations of factors affecting accident risk at both segment and zone granularities using multi-level, multi-view spatial representation blocks and temporal representation blocks. We further exploit the consistency between the two granularities to improve prediction accuracy. Additionally, we introduce a Shared-Expert Module (SEM) to capture the influence of shared factors across both granularities. To effectively address the zero-inflation issue, we design a Zero-Inflated Mixture Poisson (ZIMP) decoder that directly models the sparse distribution of accident risk data and generates the final predictions. In summary, the key contributions of this work are as follows:

- We introduce a novel framework for traffic accident analysis by incorporating a spatial-temporal knowledge graph. Building upon this idea, we propose STKGRisk, a model that simultaneously predicts traffic accident risks at both the road segment and taxi zone granularities from multi-level and multi-view perspectives.
- We construct a traffic accident spatial-temporal knowledge graph and design a diachronic embedding module to capture the high-order, dynamic interactions between multi-source factors and traffic accidents.
- We introduce a novel Zero-Inflated Mixture Poisson decoder to model the underlying distribution of sparse accident risk data, effectively mitigating the zero-inflation issue and significantly enhancing prediction performance.
- STKGRisk is evaluated on three real-world, multi-granularity traffic accident risk datasets. Experimental results demonstrate its effectiveness in predicting accident risk at both segment and zone granularities, as well as in identifying high-risk regions.

II. Related Work

A. Traffic Accident (Risk) Prediction

Urban traffic accident prediction is a critical research area in intelligent transportation systems. Existing approaches can be broadly categorized into traditional machine learning methods and deep learning methods. Representative traditional methods

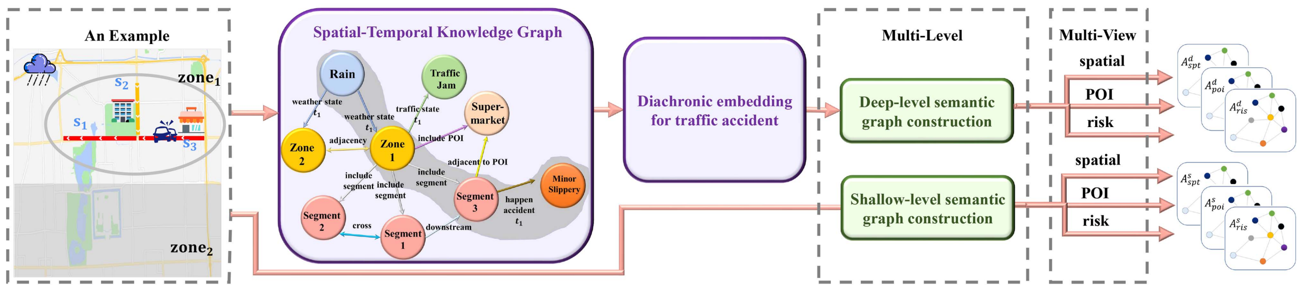


Fig. 2. A sample of a spatiotemporal traffic accident knowledge graph (STKG) that captures the complex interactions among traffic accidents and their influencing factors.

include Support Vector Machines (SVM), negative binomial regression [12], and decision trees [13]. However, these approaches generally assume that traffic accidents occur independently, an assumption that is often violated in real-world scenarios. Furthermore, they typically overlook the impact of external influencing factors, resulting in limited predictive performance.

To this end, many researchers have proposed deep learning-based approaches for traffic accident prediction. For instance, SDAE [1] and SDCAE [2] employ stacked denoising autoencoders to reconstruct traffic accident risk. However, these methods fail to capture the temporal correlations inherent in accident data. C-ViT [14] utilizes a vision transformer to capture the spatial-temporal correlations of grid-based accident risk. Recently, Spatial-temporal graph convolutional networks, which can simultaneously capture the spatial-temporal correlations of irregular traffic accident risk data, have been widely applied in traffic accident risk prediction, e.g., RiskOracle [3], GSNet [4], DSTGCN [15], HintNet [16], STZINB-GNN [17], PST-CGCN [18]. To better understand the patterns of traffic accidents and mitigate the zero-inflation issue, recent studies have explored various technical approaches and integrated diverse multi-source external information. RiskSeq [5], MVMT-STN [9], and RiskContra [10] leverage multi-granularity correlations in traffic risk. DF-TAR [6] and MG-TAR [7] further incorporate private dangerous driving records, while MGHSTN [8] integrates remote sensing data. However, these models have two major limitations. First, these models make insufficient use of external information and fail to capture the dynamic interactions among different external factors, which are critical for improving the accuracy of traffic data forecasting [19]. While EMG-GAT [20] introduces temporal knowledge graphs to incorporating different external factors for traffic risk prediction, its utilization remains limited to feature-level enhancement, lacking explicit modeling of dynamic temporal evolution and high-order relational dependencies, which restricts its ability to fully capture complex spatiotemporal interactions. Second, they cannot fundamentally address the zero-inflation issue, as they rely solely on data augmentation or the incorporation of additional external data, without explicitly modeling the underlying distribution of sparse accident risk data. Although STZINB-GNN [17] employs a zero-inflated negative binomial distribution to address sparsity, its single-distribution formulation is insufficient to model the diverse underlying generation patterns of traffic accidents.

B. Temporal Knowledge Graph Embedding

A Temporal Knowledge Graph (TKG) is essentially a dynamic heterogeneous directed graph with multiple types of entities and relations. TKGs offer strong representational power, as they can capture high-order temporal and dynamic interactions among entities. TKG embedding refers to the process of learning low-dimensional representations of entities and relations, where the learned embeddings effectively preserve the intrinsic properties of the TKG. Early TKG embedding methods include TAE [21], HyTE [22], and ConT [23]. However, these methods only learn embeddings for historical timestamps and cannot generalize to future ones. Recently, significant progress has been made in temporal knowledge graph embedding. DE-SimplE [24] introduces diachronic embeddings, which inductively compute dynamic entity embeddings given a specific timestamp. TG-Former [25] introduces a graph transformer framework to capture complex structural and temporal dependencies in knowledge graphs. DV-TKR [26] models temporal dynamics from complementary perspectives to enhance reasoning capability. HGE [27] embeds temporal knowledge graphs into a product space of heterogeneous geometric subspaces, enabling more expressive representation of diverse relational patterns. Despite their strong performance in knowledge representation and reasoning tasks, these methods are primarily designed for link prediction or reasoning, and do not directly address downstream spatiotemporal prediction problems with complex multi-source interactions. Inspired by these advances, we construct a novel Spatial-Temporal Knowledge Graph (STKG) for traffic accident data. By leveraging an embedding-based framework, our approach captures the interactions among diverse entities, enabling a more comprehensive understanding of the complex scenarios underlying traffic accidents.

III. Preliminaries

In this section, we will introduce essential definitions concerning the prediction of traffic accident risk.

Definition 1 (Traffic Accident Spatial Temporal Knowledge Graph): A Spatial Temporal Knowledge Graph (STKG) for traffic accident is denoted as $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T}, \mathcal{F})$, where $\mathcal{E}$, $\mathcal{R}$,

$\mathcal{T}$ and $\mathcal{F}$ stand for the sets of entities, relations, timestamps and quadruples. In this context, a quadruple $(h, r, u, t) \in \mathcal{F}$ is a fact that denotes a relation $r \in \mathcal{R}$ exists from the head entity $h \in \mathcal{E}$ to the tail entity $u \in \mathcal{E}$ at timestamp $t \in \mathcal{T}$.

Definition 2 (Multi-Granularity): We aim to predict the traffic accident risk in an urban district from both fine and coarse granularities. In this paper, the fine granularity regions refer to road segments, while the coarse granularity regions refer to taxi zones, i.e., the multi-granularity set is $G = \{\text{seg}, \text{ zon}\}$. Specifically, N_{seg} and N_{zon} represent the number of road segments and taxi zones respectively in the studied urban district. Each taxi zone encompasses several road segments, and each road segment is associated with at least one taxi zone. Therefore, we use $\mathbf{S} \in \mathbb{R}^{N_{\text{zon}} \times N_{\text{seg}}}$ to signify the mapping between road segments and taxi zones.

Definition 3 (Multi-Level): We explore the correlations of traffic accident risk among regions (i.e., road segments or taxi zones) at both shallow and deep levels. Specifically, at the shallow level, there exist explicit correlations between regions' traffic accident risk. In contrast, at the deep level, the correlations between regions' traffic accident risk are implicit, influenced by multiple sources. Formally, the multi-level set is denoted as $L = \{\text{dee}, \text{ sha}\}$, where dee and sha represent the deep and shallow level, respectively.

Definition 4 (Multi-View): We analyze the correlations of traffic accident risk among regions from multiple views, including spatial view, POI view, and risk view. So, the view set is denoted as $V = \{\text{spt}, \text{poi}, \text{ris}\}$.

Definition 5 (Semantic Graph): We construct a series of semantic graphs for multiple levels L, multiple views V, and multiple granularities G. Specifically, the adjacency matrix in a particular semantic graph is represented as $\mathbf{A}^l_{v,g} \in \mathbb{R}^{N_g \times N_g}$, where $l \in L,\ v \in V,\ g \in G$. The nodes in the semantic graphs are road segments or taxi zones determined by g. There are total 12 ($=|L||V||G|$) semantic graphs.

Definition 6 (Traffic Accident Risk): Traffic accident risk quantifies the effects caused by traffic accidents. Specifically, we first classify accidents into five classes based on the number of casualties in accidents and set the accident risk values for each accident to 1, 2, 4, 8, 16 from mild to severe. To get the traffic accident risk $\tilde{\mathbf{Y}}^i_{t,g}$ of region i at time slot t for a specific granularity $g \in G$, we sum up the traffic accident risk of all the accidents that occur within region i during time slot t. Moreover, in practical situations, the traffic accident risk of region i at time slot t for a specific granularity g is not solely determined by $\tilde{\mathbf{Y}}^i_{t,g}$ but also influenced by the traffic accident risk of neighboring regions along the spatial and temporal dimensions. This is because the effects of a traffic accident spread across both the temporal and spatial dimensions. To this end, we develop **a spatial-temporal propagation strategy for accident risk**, which involves recalibrating the initial $\tilde{\mathbf{Y}}^i_{t,g}$ to $\mathbf{Y}^i_{t,g}$ in order to comprehensively reflect the effects of traffic accidents. Specifically, we propagate the accident risk in an exponential decay way along the spatial and temporal dimensions. Then, the recalibrated traffic accident risk $\mathbf{Y}^i_{t,g}$ is the sum of its own initial accident risk $\tilde{\mathbf{Y}}^i_{t,g}$ and the accident risk from its spatial and temporal neighbors. Formally,

$$\mathbf{Y}^i_{t,\text{g}} = \tilde{\mathbf{Y}}^i_{t,g} + \sum_{j \in \mathcal{N}^k_i} \sum_{k=1}^{K} \tilde{\mathbf{Y}}^j_{t,g} \text{pow}(0.5, k) + \sum_{j \in \mathcal{N}^k_i} \sum_{k=1}^{K} \tilde{\mathbf{Y}}^j_{t-1,g} \text{pow}(0.3, k), \tag{1}$$

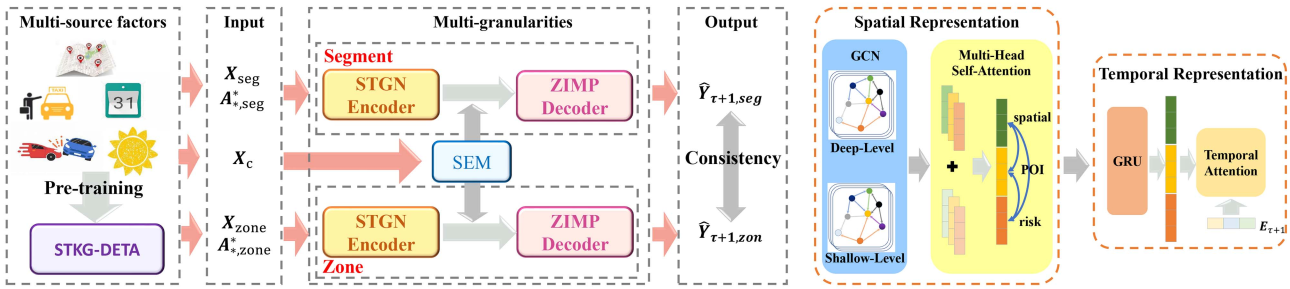
where K means the accident risk will propagate to K-order spatial neighbors at most. $\mathcal{N}^k_i$ is the set of the road segment i's kth-order neighbors. In the temporal dimension, we only consider the influences from the last time slot. Here, K is set to 3. The exponential decay parameters in spatial and temporal dimension propagation are set to 0.5 and 0.3. The primary role of this setting is to alleviate the zero-inflation issue by "spreading" the risk to neighboring spatiotemporal cells. These parameters can be flexibly adjusted according to specific regional characteristics or management requirements.

Definition 7 (Multi-Source Input): $\mathbf{X} = \{\mathbf{X}_{\text{seg}}, \mathbf{X}_{\text{zon}}, \mathbf{X}_c\}$ is the multi-source input to our model. Here, $\mathbf{X}_{\text{seg}}$ and $\mathbf{X}_{\text{zon}}$ respectively refer to the private features owing to each granularity, including the distribution of POI, the risk of accidents, and the traffic flow of the segments and taxi zones. In $\mathbf{X}_{\text{seg}}$ and $\mathbf{X}_{\text{zon}}$, the feature of POI distributions are static, while other features are dynamic. $\mathbf{X}_c$ refers to the shared features among the two granularities, including time information, weather conditions, and accident causes, which are all dynamic. Each dynamic feature comprises $T = p + q$ historical time slots, encompassing short-term and long-term features. The short-term features pertain to the characteristics of the most recent p consecutive time slots, whereas long-term features relate to those from the corresponding time slots in the previous q weeks.

Definition 8 (Problem Statement: Multi-granularity Accident Risk Prediction): We aim to jointly predict the traffic accident risk of both the fine and coarse granularities at the next time slot $\mathbf{Y}_{\tau+1,\text{seg}}$ and $\mathbf{Y}_{\tau+1,\text{zon}}$ when given the model input $\mathbf{X}$ containing features until τ.

IV. Methodology

The core idea of our proposed model, STKGRisk, is to introduce a STKG to comprehensively describe the complex and high-order interactions among multi-source entities involved in our study, enabling a dynamic representation of these entities. Building on this foundation, our model can capture the deep-level implicit correlations of traffic accident risk between regions influenced by multiple sources. The overall architecture of STKGRisk is shown in Fig. 3(a). It is made up of four modules, i.e., a spatial temporal knowledge graph diachronic embedding module (STKG-DETA), a spatial temporal graph network encoder (STGN), a shared-expert module (SEM) and a zero-inflation mixture Poisson decoder (ZIMP). Specifically, STKG-DETA module generates diachronic embeddings for entities involved in our study through pre-training the traffic accident spatial temporal knowledge graph. These embeddings serve for constructing the deep-level semantic graphs later. STGN encoder takes the private features of each granularity as input, capturing



(a) Model Architecture (b) STGN Encoder

Fig. 3. (a) shows the overall architecture of STKGRisk. It is made up of four modules, i.e., a spatial temporal knowledge graph diachronic embedding module (STKG- DETA), a spatial temporal graph network (STGN) encoder, a shared-expert module (SEM) and a zero-inflation mixture Poisson (ZIMP) decoder. (b) shows the workflow of STGN encoder. It is made up of a spatial representation block and a temporal representation block.

TABLE I
ENTITY TYPE TABLE

Id	Entity type	Number (Bro/Man/Bronx)
1	Accident Level & Cause	212/204/212
2	Traffic State Level	5/5/5
3	Taxi Service Zone	61/69/43
4	POI	5,587/5,866/3,649
5	Weather	25/25/25
6	Road Segment	1,049/475/712

TABLE II
FACT QUADRUPLE TABLE

Id	quadruple
1	(Zone, Weather State, Weather, t)
2	(Zone, Traffic State, Traffic State Level, t)
3	(Zone, Adjacency, Zone, t)
4	(Segment, Downstream, Segment, t)
5	(Segment, Cross, Segment, t)
6	(Zone, Include POI, POI, t)
7	(Segment, Adjacent to POI, POI, t)
8	(Zone, Include Segment, Segment, t)
9	(Segment, Happen Accident, Accident Level & Cause, t)

the spatial-temporal correlations of traffic accident risk between regions with the help of multi-level and multi-view semantic graphs. Besides, SEM module processes the granularity-shared features through a One-gate Mixture-of-Experts (OMoE) network [28]. Finally, ZIMP Decoder makes traffic risk predictions by adopting the ZIMP distribution to describe the patterns of sparse traffic accidents.

A. *Spatial Temporal Knowledge Graph Diachronic Embedding Module for Traffic Accident*

We propose the first spatial temporal knowledge graph (STKG) tailored for traffic accident analysis. STKG is constructed exclusively from the training dataset and can effectively capture the complex and high-order interactions between entities involved in traffic accidents. Then, we design a diachronic embedding module (DETA), which is pre-trained to generate diachronic embeddings for entities by mining the high-order relations among multi-source factors.

1) STKG for Traffic Accident: In this section, we will explain how to construct a STKG to fully depict the dynamic and complicated scenarios when traffic accidents happen. In our STKG, $\mathcal{E}$ has 6 entity types, $\mathcal{R}$ has 9 relation types, $\mathcal{F}$ has 9 kinds of fact quadruples, $\mathcal{T}$ includes all the minute-level timestamps in the training dataset. Table I shows the entity types in STKG and their corresponding numbers in our three experimental datasets (i.e., Brooklyn, Manhattan, and Bronx datasets). Table II presents the relation types and their corresponding fact quadruples in the STKG.

Our proposed traffic accident STKG can comprehensively describe the high-order spatial-temporal interactions among multiple sources when the traffic accident occurs. For example, Fig. 2 presents a sample of traffic accident STKG, where different colored circles represent different entity types and different colored edges with directions represent different relation types. We can observe that the multi-step shadowed path in STKG is able to embody the high-order interactions among weather, taxi zones, segments and traffic accidents.

2) Diachronic Embedding for Traffic Accident: Considering the dynamics of entities in STKG, we design a diachronic embedding module (DETA) to obtain the embeddings for entities at each minute-level timestamp. DETA module consists of a relation embedding, a diachronic entity embedding, and a score function.

Specifically, the embeddings of relation $r \in \mathcal{R}$ can be denoted as $\{\mathbf{f}_r, \mathbf{f}_{\bar{r}}\}$, where $\mathbf{f}_r \in \mathbb{R}^d$ is a learnable vector of relation r, while $\mathbf{f}_{\bar{r}} \in \mathbb{R}^d$ is a learnable vector of reverse relation $\bar{r}$.

For an entity h as the head entity, its diachronic entity embedding at timestamp t consists of a dynamic part and a static part. Formally, it is denoted as:

$$\mathbf{z}_h^t[i] = \begin{cases} \sum_{j \in P_t} \mathbf{a}_h^j[i] \sin(\mathbf{w}_h^j[i] j + \mathbf{b}_h^j[i]), & 1 \leq i \leq \gamma d, \\ \mathbf{m}_h[i], & \gamma d < i \leq d, \end{cases} \tag{2}$$

where γ is the ratio of dynamic part. P_t is a temporal information set containing four types of time features i.e., year, month, day and minute-level timestamp. $\mathbf{a}_h^j$, $\mathbf{w}_h^j$, $\mathbf{b}_h^j \in \mathbb{R}^{\gamma d}$ are learnable parameters for time feature type j. $\mathbf{m}_h \in \mathbb{R}^{d-\gamma d}$ represents the static part which are learnable. $\dot{\mathbf{z}}_h^t$ refers to the embedding of entity h as tail entity at timestamp t, which has the same definition with $\mathbf{z}_h^t$ but possesses different learnable parameters.

Given arbitrary fact quadruple $f = (h, r, u, t)$, score function denotes the probability of the fact is hold. It is computed by the average of matching scores of the fact quadruple and its corresponding reverse one. Formally,

$$\psi(f) = \sum \frac{\mathbf{z}_h^t \mathbf{f}_r \dot{\mathbf{z}}_u^t + \mathbf{z}_u^t \mathbf{f}_{\bar{r}} \dot{\mathbf{z}}_h^t}{2}, \tag{3}$$

where $\mathbf{z}_u^t$ and $\dot{\mathbf{z}}_h^t$ are diachronic embeddings of the head and tail entities in the reverse fact $(u, \bar{r}, h, t)$ and the matching score increases with the similarity between the head embedding linearly mapped by the relation embedding and the tail embedding.

DETA is pre-trained on the traffic accident STKG. For a fact quadruple $f = (h, r, u, t)$ in $\mathcal{G}$, we randomly replace the head entity or tail entity with other entities to acquire negative samples, while itself is regarded as the positive sample. Then, we use the cross-entropy loss function to guide DETA to correctly distinguish positive and negative samples. Later, DETA will be further fine-tuned during the training process of the whole model. Finally, STKG-DETA generates diachronic entity embeddings reflecting dynamic high-order interactions between entities at the target time slot, which are then used to construct deep-level graphs in the following STGN encoder.

B. Spatial Temporal Graph Network Encoder

The STGN encoder aims to model the spatiotemporal correlations of traffic accident risk from multi-level semantics and multiple views for each granularity, based on the granularity-specific features $\mathbf{X}_{\text{seg}}$ and $\mathbf{X}_{\text{zon}}$. Fig. 3(b) illustrates the workflow of the STGN encoder. Since the computation processes are identical for both granularities, we take the segment level as an example for illustration. The outputs of the STGN encoders are denoted as $\mathbf{M}_{\text{seg}}$ and $\mathbf{M}_{\text{zon}}$.

1) Shallow Level Semantic Graph Construction: From the perspective of shallow-level semantics, there exist explicit correlations between the accident risk of regions (i.e., segments and zones), which can be directly calculated by the mathematical methods based on the granularity-private features. To comprehensively represent these correlations, we construct a series of shallow level semantic graphs from multiple views.

For the spatial view, we construct the shallow level semantic graphs $\mathbf{A}_{\text{spt,seg}}^{\text{sha}}$ and $\mathbf{A}_{\text{spt,zone}}^{\text{sha}}$ based on the topological connections between regions. For the POI view and the risk view, we construct the shallow level semantic graphs based on the similarities between regions regarding their feature distributions. In this paper, Jensen-Shannon divergence [3] is employed to calculate the similarity. Specifically, Jensen-Shannon divergence measures the difference between two distributions and it is symmetric. The smaller the divergence value, the smaller the difference. The feature distributions in the POI view and risk view are the POI count distributions and the accident risk sequences in training datasets within each region. Formally,

$$\mathbf{A}_{v,g}^{\text{sha}}(i,j) = 1 - \text{JS}\left(\mathbf{R}_{v,g}^i, \mathbf{R}_{v,g}^j\right),$$

$$\text{JS}(\mathbf{R}_{v,g}^i, \mathbf{R}_{v,g}^j) = \frac{1}{2} \sum_{1 \le e \le d_{js}} \Bigg(\mathbf{R}_{v,g}^i(e) \log \frac{2\mathbf{R}_{v,g}^i(e)}{\mathbf{R}_{v,g}^i(e) + \mathbf{R}_{v,g}^j(e)} + \mathbf{R}_{v,g}^j(e) \log \frac{2\mathbf{R}_{v,g}^j(e)}{\mathbf{R}_{v,g}^i(e) + \mathbf{R}_{v,g}^j(e)} \Bigg), \tag{4}$$

where $v \in \{\text{poi}, \text{ris}\}$, $g \in G$, $\mathbf{R}_{v,g}^i \in \mathbb{R}^{d_{js}}$ is the feature distribution of region i under view v. Finally, we obtain four shallow level semantic graphs, i.e., $\mathbf{A}_{\text{poi,seg}}^{\text{sha}}$, $\mathbf{A}_{\text{ris,seg}}^{\text{sha}}$, $\mathbf{A}_{\text{poi,zone}}^{\text{sha}}$, and $\mathbf{A}_{\text{ris,zone}}^{\text{sha}}$.

2) Deep Level Semantic Graph Construction: We utilize diachronic entity embeddings that capture the dynamic, high-order interactions between multi-source factors to construct deep level semantic graphs. So these deep level semantic graphs are expected to capture the implicit spatial-temporal correlations of risks between regions affected by multi-source factors.

Specifically, let $\mathbf{E}_{\tau+1}^{\text{seg}} \in \mathbb{R}^{N_{\text{seg}} \times d}$, $\mathbf{E}_{\tau+1}^{\text{zon}} \in \mathbb{R}^{N_{\text{zon}} \times d}$, $\mathbf{E}_{\tau+1}^{p} \in \mathbb{R}^{P \times d}$, $\mathbf{E}_{\tau+1}^{f} \in \mathbb{R}^{F \times d}$ denote the diachronic embedding matrices of road segment entity, taxi zone entity, POI entity, accident level & cause entity at the target time slot $\tau + 1$. P and F is the number of POI entity and accident level & cause entity respectively. Based on these diachronic embedding matrices, we construct six deep level semantic graphs from the spatial, POI and risk views for the segment and zone granularities.

For the spatial view, the deep level semantic graphs are constructed in an adaptive way,

$$\mathbf{A}_{\text{spt},g}^{\text{dee}} = \text{softmax}(\text{ReLU}(\mathbf{E}_{\tau+1}^{g} (\mathbf{E}_{\tau+1}^{g})^T)) + \mathbf{I}_g, \tag{5}$$

where $g \in G$, $\mathbf{I}_g \in \mathbb{R}^{N_g \times N_g}$ is an identity matrix.

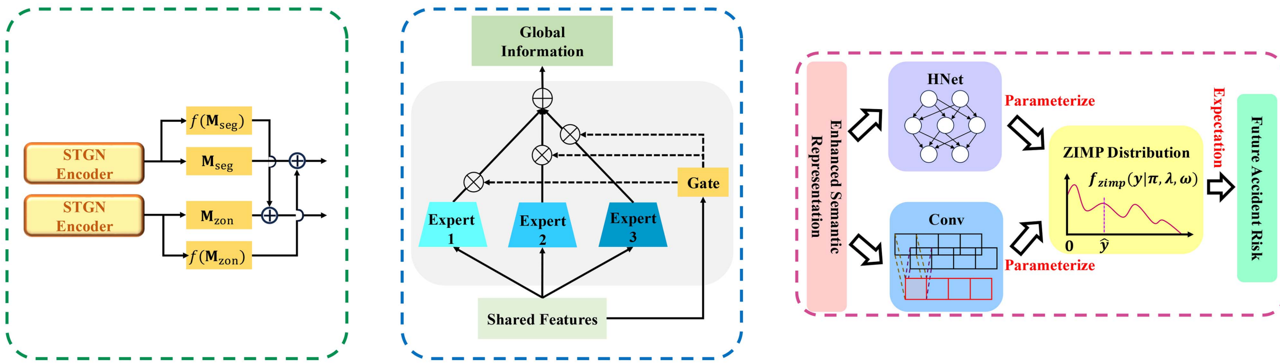
To get the deep level semantic graphs for the risk view and the POI view, we first find mapping relations between risk/POI and regions, then adopt the same adaptive way to construct graphs. Next, we take deep level semantic graph for segments in the POI view as an example to illustrate the calculation process.

Specifically, we define a transfer matrix $\mathbf{T}_{\text{seg}}^{p} \in \mathbb{R}^{N_{\text{seg}} \times P}$ to represent the mapping relations between POI and segments. The values of each element in the transfer matrix are assigned to the scores calculated by (3) with the fact quadruple (road segment, adjacent to POI, POI, τ_{train}), where τ_{train} is the latest timestamp in training dataset. Then, we can construct the graph $\mathbf{A}_{\text{poi,seg}}^{\text{dee}}$ in the adaptive way. Formally,

$$\mathbf{M}_p^{\text{seg}} = \mathbf{T}_{\text{seg}}^{p} \mathbf{E}_{\tau_{\text{train}}+1}^{p},$$

$$\mathbf{A}_{\text{poi,seg}}^{\text{dee}} = \text{softmax}(\text{ReLU}\left(\mathbf{M}_p^{\text{seg}} \left(\mathbf{M}_p^{\text{seg}}\right)^T\right) + \mathbf{I}_{\text{seg}}. \tag{6}$$

Finally, all the semantic graphs along with the granularity-private features $\mathbf{X}_{\text{seg}}$, $\mathbf{X}_{\text{zon}}$ will be fed into spatial representation blocks and temporal representation blocks for further computation. Such computation process is the same for each granularity, so we only take the fine granularity, i.e., road segment, as an example to introduce the two blocks.



(a) Message Sharing (b) SEM (c) ZIMP Decoder

Fig. 4. (a) shows the message sharing mechanism. (b) shows the workflow of Shared-Expert Module (SEM). (c) shows the workflow of Zero-inflation Mixture Poisson (ZIMP) decoder.

3) Spatial Representation Block: We employ two layers of graph convolution networks (GCN) to aggregate and propagate the granularity-private features $\mathbf{X}_{\text{seg}}$ along the spatial dimensions based on the semantic graphs. Formally,

$$\mathbf{Z}_v^l = \text{ReLU}(\mathbf{A}_{v,\text{seg}}^l \text{ReLU}(\mathbf{A}_{v,\text{seg}}^l \mathbf{X}_{\text{seg}} \mathbf{W}_v^l + \mathbf{b}_v^l)\mathbf{U}_v^l + \mathbf{c}_v^l), \tag{7}$$

where $v \in V, l \in L$, $\mathbf{W}_v^l$, $\mathbf{b}_v^l$, $\mathbf{U}_v^l$ and $\mathbf{c}_v^l$ are learnable matrices, $\mathbf{Z}_v^l \in \mathbb{R}^{T\times N_{\text{seg}}\times D}$, D is the kernel size of GCN. For each view, we fuse the output based on the shallow and deep semantic graphs in an adaptive weighted way,

$$\mathbf{Z}_v = c_v \mathbf{Z}_v^{\text{sha}} + \mathbf{Z}_v^{\text{dee}}, \tag{8}$$

where c_v is a learnable parameter whose initial value is 0.5. Then we concatenate the output of different view $\mathbf{Z} = \mathbf{Z}_{\text{spt}}||\mathbf{Z}_{\text{poi}}||\mathbf{Z}_{\text{ris}} \in \mathbb{R}^{T\times N_{\text{seg}}\times 3D}$.

The above GCNs independently take the local spatial correlations in different semantic aspects into considerations. In order to further capture the global spatial correlation between regions, we perform the Multi-Head Attention (MHA) layers [29] on $\mathbf{Z}$ to obtain $\mathbf{Z}' \in \mathbb{R}^{T\times N_{\text{seg}}\times 3D}$.

4) Temporal Representation Block: After modeling the spatial correlation of $\mathbf{X}_{\text{seg}}$, we further utilize Gated Recurrent Unit (GRU) to capture its temporal correlations. Since the historical data at different historical time slots have varying effect on the target time slot, so we introduce the temporal attention mechanism to aggregate historical data.

$$\begin{aligned} \bar{\mathbf{Z}} &= \text{GRU}\left(\mathbf{Z}'\right), \\ \boldsymbol{\omega} &= \text{softmax}\left(\text{ReLU}\left(\bar{\mathbf{Z}}\mathbf{W}_t + \mathbf{E}_{\tau+1}\mathbf{W}_e + \mathbf{b}_a\right)\right), \end{aligned} \tag{9}$$

where $\bar{\mathbf{Z}} \in \mathbb{R}^{T\times N_{\text{seg}}\times d_h}$, d_h is the hidden size of GRU. $\mathbf{W}_e$, $\mathbf{W}_t$ and $\mathbf{b}_a$ are learnable parameters, $\mathbf{E}_{\tau+1}$ is the time embedding of target time slot $\tau+1$, $\boldsymbol{\omega} \in \mathbb{R}^T$ represent the attention score of each time slot. Finally, the output of the temporal representation block is $\mathbf{M}_{\text{seg}} = \sum_{i=1}^{T} \boldsymbol{\omega}_i \bar{\mathbf{Z}}_i \in \mathbb{R}^{N_{\text{seg}}\times d_h}$.

5) Message Sharing: A message sharing mechanism between fine and coarse granularities in latent space is further employed, which contributes to keeping the message consistency between multi-granularity. As shown in Fig. 4(a), it is defined as follow where $\mathbf{S} \in \mathbb{R}^{N_{\text{zon}}\times N_{\text{seg}}}$ signifies the mapping between road segments and taxi zones,

$$\begin{aligned} \mathbf{M}_{\text{seg}} &= \mathbf{M}_{\text{seg}} + \text{ReLU}\left(\mathbf{S}^T \mathbf{M}_{\text{zon}} \mathbf{W}_{\text{seg}}\right), \\ \mathbf{M}_{\text{zon}} &= \mathbf{M}_{\text{zon}} + \text{ReLU}\left(\mathbf{S}\mathbf{M}_{\text{seg}} \mathbf{W}_{\text{zon}}\right). \end{aligned} \tag{10}$$

C. Shared-Expert Module (SEM)

The granularity-shared features $\mathbf{X}_c$ in both fine and coarse granularities contain multiple kinds of global semantic information which are valuable for accident risk prediction. Since these multiple kinds of information within $\mathbf{X}_c$ have different characteristics, we utilize OMoE to deal with $\mathbf{X}_c$, expecting each expert is responsible for handling a specific part of features. The workflow diagram are shown in Fig. 4(b).

$$\begin{aligned} \Phi(\mathbf{X}_c) &= \text{softmax}(\mathbf{W}_g \mathbf{X}_c), \\ \hat{\mathbf{M}}_{\text{seg}} &= \mathbf{M}_{\text{seg}} + \sum_{j=1}^{J} \Phi(\mathbf{X}_c)_j \mathbf{V}^j \mathbf{X}_c, \\ \hat{\mathbf{M}}_{\text{zon}} &= \mathbf{M}_{\text{zon}} + \sum_{j=1}^{J} \Phi(\mathbf{X}_c)_j \mathbf{V}^j \mathbf{X}_c, \end{aligned} \tag{11}$$

where $\mathbf{V}^j$ is the jth expert network implemented by Multiple Layer Perceptrons (MLPs), J is the number of expert networks, Φ is a gated network. Finally, we get the enhanced semantic representations $\hat{\mathbf{M}}_{\text{seg}} \in \mathbb{R}^{N_{\text{seg}}\times d_h}$, $\hat{\mathbf{M}}_{\text{zon}} \in \mathbb{R}^{N_{\text{zon}}\times d_h}$.

D. Zero-Inflation Mixture Poisson (ZIMP) Decoder

Considering the zero-inflation issue of sparse accident data, we introduce zero-inflation Poisson distribution [30], [31], [32] to analyze traffic accident data. Since different regions show heterogeneity in the occurrence patterns of traffic accidents which cannot be adequately captured by a single Poisson distribution, we innovatively propose zero-inflation Mixture Poisson (ZIMP) to describe accident risk data.

Mathematically, the probability mass function (PMF) of ZIMP to describe the traffic accident risk y of a region can be written as:

$$f_{\text{zimp}}(y) = \begin{cases} \pi + (1-\pi)\sum_{i=1}^{m} w_i e^{-\lambda_i}, & y = 0, \\ (1-\pi)\sum_{i=1}^{m} w_i \frac{\lambda_i^y}{y!} e^{-\lambda_i}, & y > 0, \end{cases} \tag{12}$$

where π is the sparsity parameter that refers to the probability of zero value in data, m is the number of mixture components. Each component is independent Possion distribution with different intensity parameters λ_i. All components are combined with weights w_i.

We parameterize the parameter π, w and λ by decoding the enhanced semantic representation as shown in Fig. 4(c). We take the fine granularity as an example, the computation processes are as follows,

$$\begin{aligned} \boldsymbol{\pi} &= \text{sigmoid}(\text{Conv}(\hat{\mathbf{M}}_{\text{seg}})), \\ \boldsymbol{\lambda} &= \text{softplus}(\text{HNet}(\hat{\mathbf{M}}_{\text{seg}})), \\ \boldsymbol{w} &= \text{softmax}(\text{HNet}(\hat{\mathbf{M}}_{\text{seg}})), \end{aligned} \tag{13}$$

where $\boldsymbol{\pi} \in \mathbb{R}^{N_{\text{seg}}}$, $\boldsymbol{\lambda} \in \mathbb{R}^{N_{\text{seg}} \times m}$, $\boldsymbol{w} \in \mathbb{R}^{N_{\text{seg}} \times m}$. Conv is a CNN with kernel size 1. HNet consists of MLP and activation function Tanh. Finally, We predict the future accident risk by calculate the expectation of ZIMP Distribution,

$$\hat{\mathbf{Y}}_{\tau+1,\text{seg}} = (1-\boldsymbol{\pi})\sum_{i=1}^{m} \boldsymbol{w}_i \boldsymbol{\lambda}_i. \tag{14}$$

E. Loss Function

The loss functions $\mathcal{L}_{\text{seg}}$ and $\mathcal{L}_{\text{zon}}$ for accident risk predictions at the fine granularity and coarse granularity have the same format. We take the fine granularity accident risk prediction as an example to illustrate the loss function. Formally,

$$\mathcal{L}_{\text{seg}} = (1-a)\mathcal{L}_{\text{nll}} + a\mathcal{L}_{\text{mse}}, \tag{15}$$

where a is a hyperparameter to balance the two terms in loss. $\mathcal{L}_{nll}$ is the negative log-likelihood of ZIMP Distribution, which is designed to effectively guide model to learn the distribution parameters. $\mathcal{L}_{mse}$ is a weighted MSE [4] to guide the model to make accurate predictions.

To further preserve the consistency between fine and coarse granularity accident risk predictions, we introduce consistency loss,

$$\mathcal{L}_{\text{con}} = \frac{\sum(\hat{\mathbf{Y}}_{\tau+1,\text{seg}}\mathbf{S}^T - \hat{\mathbf{Y}}_{\tau+1,\text{zon}})^2}{N_{\text{zon}}}, \tag{16}$$

where $\hat{\mathbf{Y}}_{\tau+1,\text{seg}}$ and $\hat{\mathbf{Y}}_{\tau+1,\text{zon}}$ are the prediction values of fine and coarse granularity respectively.

Finally, the total loss is expressed as the sum of the above three terms,

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \mathcal{L}_{\text{zon}} + \mathcal{L}_{\text{con}}. \tag{17}$$

V. Experiments

To evaluate the performance of our model, we conducted extensive experiments on three real-world traffic accident datasets. The implementation of our model is publicly available at https://github.com/guoshnBJTU/STKGRisk.

TABLE III
Statistics of Datasets

Dataset	Brooklyn	Manhattan	Bronx
#Accidents	41,167	28,287	22,183
#POIs	5,587	5,866	3,649
#Taxi orders	6,259k	80,184k	2,373k
#Hours of weather	8,760	8,760	8,760
#Segments	1,049	475	712
#Zones	61	69	43

A. Dataset

We collect urban data from New York City[1], including information on road segments, taxi zones, traffic accidents, weather, POI, and taxi orders. After preprocessing operations i.e., filtering and concatenation, road segment lengths range from 500 to 2000 meters. Taxi zones are defined based on the pickup and drop-off locations recorded in the taxi order data. Taxi order data is utilized to estimate traffic flow. The traffic accident dataset covers all reported accidents in New York City during 2019 and includes detailed contributing factors for each. Weather data comprises temperature and general weather conditions. POI data captures the distribution of various facilities around each road segment or within a taxi zone. Using the above open-source datasets, we construct three real-world, multi-granularity traffic accident risk datasets corresponding to the Brooklyn, Manhattan, and Bronx regions. Descriptive statistics of the datasets are summarized in Table III.

B. Settings & Baseline Models

1) Settings: The datasets are divided into training, validation and test sets at a ratio of 6:2:2 along time dimension. The granularity of time slot is an hour. For sample construction, the temporal window size T is set to 7, with $p = 3$ and $q = 4$. We implement our model and all the baselines with PyTorch.

During the pre-training phase of DETA, the hyperparameter γ is set to 0.32, the dimension of diachronic embedding d is set to 100, the number of negative samples is fixed at 500, DETA is pre-trained for 100 epochs. To train STKGRisk, we search the best hyperparameter combinations through the AutoML tool NNI. The selected hyperparameter settings are summarized in Table V.

As for metrics, we adopt Root Mean Square Error (RMSE) to measure the regression error. Inspired by related works [4], we employ Recall and Mean Average Precision (MAP) to evaluate the prediction accuracy for high accident risk regions, which is of most concern in practice. Similar to [4], RMSE, Recall and MAP can be formulated as:

$$\text{RMSE} = \frac{1}{I}\sqrt{\sum_{i=1}^{I}(\hat{\mathbf{Y}}_i - \mathbf{Y}_i)^2},$$

$$\text{Recall} = \frac{1}{I}\sum_{i=1}^{I}\frac{|R_i \cap P_i|}{|R_i|},$$

[1]https://opendata.cityofnewyork.us/

TABLE IV
COMPARISON OF MULTI-GRANULARITY PREDICTION PERFORMANCE ACROSS DIFFERENT DATASETS OVER ALL TIME SLOTS (WITH † INDICATING OUR MODELS)

Datasets	Granularity	Metrics (Baseline methods)	MLP	GRU	SDCAE	ConvLSTM	AGCRN	GSNet	MVMT-STN	MG-TAR	MGHSTN	**STKGRisk†**
Brooklyn	Road Segment	RMSE↓	0.3503	0.3436	0.3027	0.3114	0.3258	0.3180	0.3369	0.3481	0.3424	**0.2677**
		Recall (%)↑	15.43	13.23	4.07	23.74	16.16	18.25	23.35	25.86	24.08	**26.48**
		MAP(%)↑	18.98	24.92	19.02	33.67	15.89	23.44	21.75	22.34	20.18	**48.94**
	Taxi Zone	RMSE↓	4.2575	4.0109	3.8649	3.7656	3.7534	3.6328	4.083	3.7716	3.7661	**3.5023**
		Recall (%)↑	47.26	44.02	41.57	53.80	53.87	52.59	**56.05**	55.23	54.04	55.95
		MAP (%)↑	57.74	52.30	58.87	57.42	60.61	61.36	57.09	57.20	58.42	**76.61**
Manhattn	Road Segment	RMSE↓	0.3085	0.3466	0.3009	0.3139	0.3823	0.3395	0.3457	0.3542	0.3155	**0.2590**
		Recall (%)↑	10.69	6.34	2.66	15.25	16.18	16.13	16.71	17.61	**23.89**	19.13
		MAP (%)↑	23.41	15.09	15.01	28.31	11.73	24.56	18.22	25.71	29.20	**36.77**
	Taxi Zone	RMSE↓	2.8972	3.0016	2.9418	2.7295	2.9299	2.7628	3.1628	2.8085	2.8381	**2.6502**
		Recall (%)↑	48.95	44.00	42.56	52.36	50.01	54.32	50.17	49.87	48.18	**56.64**
		MAP (%)↑	61.68	58.07	49.79	67.27	61.30	66.67	59.61	60.14	52.59	**80.56**
Bronx	Road Segment	RMSE↓	0.3533	0.3080	**0.2757**	0.3000	0.3120	0.3050	0.2875	0.3354	0.3428	0.3099
		Recall (%)↑	32.49	32.36	27.12	35.32	38.47	37.89	40.50	41.41	37.21	**45.85**
		MAP (%)↑	19.97	21.17	29.99	22.86	18.99	24.31	25.02	24.20	21.86	**34.99**
	Taxi Zone	RMSE↓	4.3623	3.3181	3.2716	3.3644	3.1864	3.1599	3.1584	3.3463	3.1754	**3.0759**
		Recall (%)↑	40.20	59.80	51.35	61.79	56.75	62.03	59.64	**65.18**	60.54	63.25
		MAP (%)↑	18.59	33.17	24.92	31.51	49.98	45.62	45.11	43.40	45.78	**51.46**

$$\text{MAP} = \frac{1}{I}\sum_{i=1}^{I}\frac{\sum_{j=1}^{L} Pre(j) \times idx(j)}{|R_i \cap P_i|}, \tag{18}$$

where I is the number of samples. For the sample i, $\mathbf{Y}_i$ is the true accident risk values of all the regions at time slot i and $\hat{\mathbf{Y}}_i$ is predicted accident risk values. R_i is a set consisting of true high-risk regions and P_i is a set consisting of top L highest predictive risk regions. $Pre(j)$ represents the precision of a cut-off rank list from 1 to j, $idx(j)$ is an indicator when the jth predictive high risk region is the true high risk region, its value is 1, otherwise 0. L is set to 50 for fine granularity and 20 for coarse granularity. RMSE focuses on the prediction accuracy, the smaller the better, while Recall and MAP focus on the recall rate and average precision of high accident risk regions, the larger the better. The above RMSE, Recall and MAP are first used to measure the model performance on all time slots. To obtain more comprehensive comparisons, we also evaluate the prediction performance over the high-risk time slots which is also of most concern in practice, and the corresponding metrics are termed as RMSE*, Recall*, MAP*. According to statistical analysis on the training dataset, the high-risk hours set is {14, 15, 16, 17, 18, 19}.

2) Baseline Models: We select nine baseline models, including both accident risk forecasting methods and representative traffic forecasting models, as the study on traffic forecasting [33], [34], [35], [36] aims to effectively capture complex spatiotemporal dependencies that are also essential for risk prediction tasks.

- *MLP* consists of multiple fully connected layers with activation functions.
- *GRU [37]* captures the complex temporal dependencies.
- *ConvLSTM [38]* captures the spatio-temporal correlations by fusing convolutions and LSTM.
- *SDCAE [2]* designs a stack denoise convolutional autoencoder to capture spatial dependencies.
- *AGCRN [33]* combines adaptive GCN and GRU to capture spatio-temporal correlations of traffic data.
- *GSNet [4]* proposes a framework to capture complex spatio-temporal correlations from both geographical and semantic aspects for traffic accident risk prediction.
- *MVMT-STN [9]* proposes a multi-task framework to forecast the fine and coarse granularity traffic accident risk simultaneously.
- *MG-TAR [7]* exploits dangerous driving data and multi-view graph learning to achieve accident risk prediction. Due to the lack of dangerous driving data, in our study we only use traffic flow to construct traffic pattern graph.
- *MGHSTN [8]* proposes a novel multi-granularity hierarchical spatio-temporal network and incorporates remote sensing data to build hierarchical multi-granularity structure. Due to the lack of remote sensing data, in our study we omit this part.

C. Experimental Results

Tables IV and VI present the experimental results across the three datasets. For clarity, the best performance for each metric is shown in **bolded**, while the second-best results are underlined. As observed, STKGRisk consistently achieves the best overall performance across nearly all six evaluation metrics, spanning both granularity levels and all three datasets. Notably, the improvements in MAP and MAP* metrics are particularly significant, indicating that STKGRisk effectively identifies high-risk regions with high precision.

MLP fails to effectively capture spatiotemporal correlations of traffic accident risk. GRU focuses solely on temporal dependencies while neglecting spatial correlations, resulting

TABLE V
HYPERPARAMETER SETING

Hyperparameter	Brooklyn	Manhattan	Bronx
learning rate	0.0005	0.001	0.0001
batch size	3	8	3
GRU layers	1	1	1
GRU hidden size	128	64	16
GCN kernel size	16	32	16
MHA layers	3	2	1
Attention heads	8	4	6
Expert networks	64	4	16
a	0.5	0.4	0.7
m	128	64	256

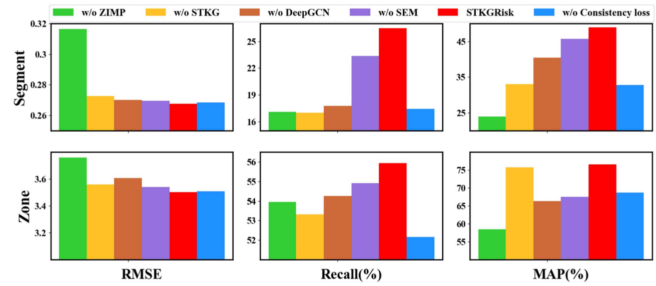


Fig. 5. Visualization of ablation experiment results.

in suboptimal performance. SDCAE leverages convolutional operations to model spatial dependencies but overlooks temporal dynamics, which also limits its effectiveness.

ConvLSTM significantly improves performance over GRU and SDCAE by jointly modeling spatial and temporal dependencies. AGCRN introduces adaptive modules to learn latent traffic accident patterns. However, it relies on a single semantic view and thus cannot capture more comprehensive patterns across multiple semantics.

GSNet and MG-TAR improve upon AGCRN by constructing multi-view graph structures to better capture spatiotemporal correlations. Building on this, MVMT-STN further incorporates spatial associations across fine and coarse granularities, enabling multi-granularity accident risk prediction. MGHSTN the latest baseline model demonstrates satisfactory performance, even the best in some cases. However, its performance lacks stability and it does not perform well on RMSE.

However, all these baselines are limited in that they primarily model the correlations of traffic accident risk from a shallow semantic level, failing to capture the implicit, dynamic and high-order correlations that could further enhance prediction accuracy. Additionally, while some models, such as GSNet, attempt to address the zero-inflation nature of accident risk data by incorporating specialized loss functions, they do not directly model the underlying distribution of the data. As a result, they are unable to tackle the data sparsity issue from a fundamental perspective. In contrast, STKGRisk explicitly learns the distribution of accident risk data through a zero-inflation mixture Poisson distribution, which allows it to capture traffic accident patterns and precisely identify high-risk regions with greater accuracy.

D. Ablation Study

We design the following four variants of STKGRisk to demonstrate the effectiveness of each part:

- *w/o ZIMP:* We replace the ZIMP decoder with a normal decoder implemented by MLP and only use weighted MSE loss.
- *w/o STKG:* We remove the STKG-DETA and replace the diachronic embedding with learned embeddings with random initialization.
- *w/o DeepGCN:* We remove DeepGCN but keep ShallowGCN in STGN Encoder.
- *w/o SEM:* We remove the SEM module, and the granularity-shared features are concatenated with granularity-private features as model input.
- *w/o consistency loss:* We remove the consistency loss in (17).

We perform ablation experiments on the Brooklyn dataset, with results presented in Fig. 5. The first row of figures shows the experimental results for segment-level accident risk prediction, while the second row presents the results for zone-level accident risk prediction. The histograms in each of the three columns display the prediction performance of all models across three metrics: RMSM, Recall, and MAP, respectively.

As observed, the first variant (without ZIMP) performs the worst overall, clearly demonstrating that modeling the potential distribution of accident risk data using the ZIMP distribution is both useful and essential. Furthermore, we find that all the designs, including STKG, DeepGCN, SEM, and consistency loss, have a positive effect on performance.

E. Case Study

To further illustrate the prediction performance of our model, we visualize both the ground truth and predicted values on the Brooklyn dataset. Specifically, we randomly select 24 road segments at the fine-granularity level and 10 taxi zones at the coarse-granularity level, using data from two separate days for each granularity. Figs. 6 and 7 show the comparison for segment-granularity and zone-granularity, respectively. In each heatmap, the horizontal axis represents time slots, and the vertical axis corresponds to either road segments or taxi zones. Each row displays the comparison between ground truth and prediction for a single day (i.e., 24 time slots). Darker colors indicate higher accident risk values.

We find that the ground truth accident risk data is highly sparse, particularly at the segment level, leading many predicted values to tend toward zero. Despite a few numerical differences between the ground truth and predictions, many high-risk road segments and taxi zones are correctly assigned relatively high predicted risk values. This indicates that our model can effectively identify high-risk areas in advance. Such predictive insights are often more important and valuable in practice than

TABLE VI
COMPARISON OF MULTI-GRANULARITY PREDICTION PERFORMANCE ACROSS DIFFERENT DATASETS OVER HIGH-RISK TIME SLOTS (WITH † INDICATING OUR MODELS)

Baseline methods			MLP	GRU	SDCAE	ConvLSTM	AGCRN	GSNet	MVMT-STN	MG-TAR	MGHSTN	**STKGRisk†**
Datasets	Granularity	Metrics										
Brooklyn	Road Segment	RMSE↓	0.3778	0.3602	0.3553	0.3536	<u>0.3495</u>	0.3559	0.3569	0.3627	0.3746	**0.3132**
		Recall (%)↑	17.13	13.70	8.78	<u>28.40</u>	15.80	22.36	22.60	25.06	23.47	**29.81**
		MAP(%)↑	24.79	30.37	30.16	<u>40.01</u>	20.08	28.24	26.79	28.24	25.15	**56.90**
	Taxi Zone	RMSE↓	5.0192	4.5004	4.5295	4.2959	4.2823	<u>4.1691</u>	4.3724	4.2841	4.2474	**4.1378**
		Recall (%)↑	44.42	43.59	38.66	53.34	53.32	<u>54.07</u>	53.41	53.14	52.21	**59.95**
		MAP (%)↑	67.27	63.58	69.19	69.68	72.10	<u>72.59</u>	68.86	68.53	70.01	**81.47**
Manhattn	Road Segment	RMSE↓	0.3579	0.3679	0.3507	0.3526	0.3982	0.3704	0.3675	0.3667	<u>0.3504</u>	**0.3076**
		Recall (%)↑	16.11	5.85	5.78	20.48	15.34	20.63	14.38	17.59	<u>22.62</u>	**24.72**
		MAP (%)↑	32.74	18.86	25.44	36.04	14.43	27.79	22.68	32.05	<u>35.58</u>	**49.38**
	Taxi Zone	RMSE↓	3.4290	3.4165	3.4908	<u>3.2334</u>	3.4200	3.2854	3.3547	3.2549	3.2469	**3.1395**
		Recall (%)↑	47.15	44.97	41.81	53.08	49.33	52.97	50.82	50.70	<u>57.36</u>	**58.66**
		MAP (%)↑	72.93	69.43	59.60	76.18	71.80	<u>76.71</u>	71.32	71.76	61.38	**85.28**
Bronx	Road Segment	RMSE↓	0.3458	0.3328	<u>0.3274</u>	0.3484	0.3368	0.3462	**0.3215**	0.3524	0.3719	0.3595
		Recall (%)↑	33.06	32.16	36.54	36.57	35.95	36.43	<u>40.4</u>	39.6	36.13	**48.24**
		MAP (%)↑	25.27	26.45	<u>32.03</u>	27.82	24.08	29.46	30.56	30.06	26.17	**43.06**
	Taxi Zone	RMSE↓	4.4992	3.6423	3.6949	3.817	3.8548	**3.5567**	3.7364	3.679	<u>3.6135</u>	3.6644
		Recall (%)↑	38.78	57.38	52.97	58.41	64.36	67.18	<u>69.34</u>	68.07	65.81	**71.09**
		MAP (%)↑	22.98	41.96	31.06	38.32	<u>52.92</u>	50.16	48.95	48.56	51.41	**56.01**

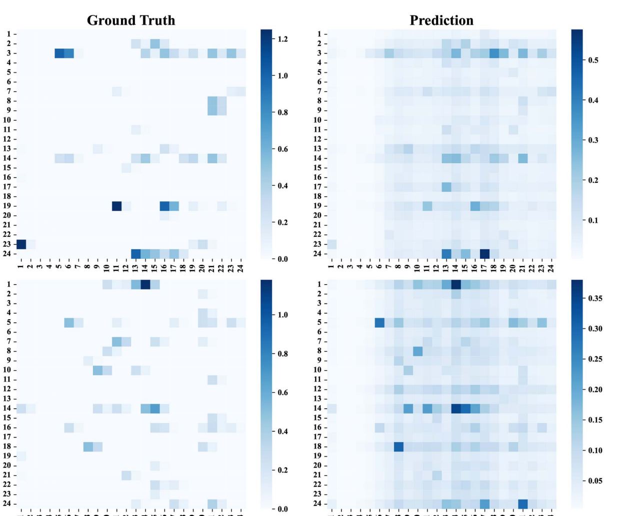


Fig. 6. Visualization of ground truth and prediction in segment-granularity accident risk prediction.

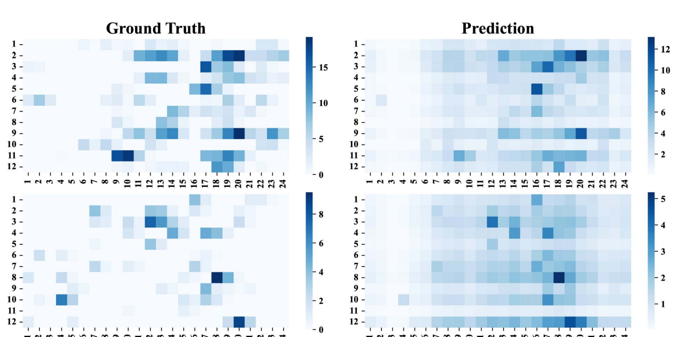


Fig. 7. Visualization of ground truth and prediction in zone-granularity accident risk prediction.

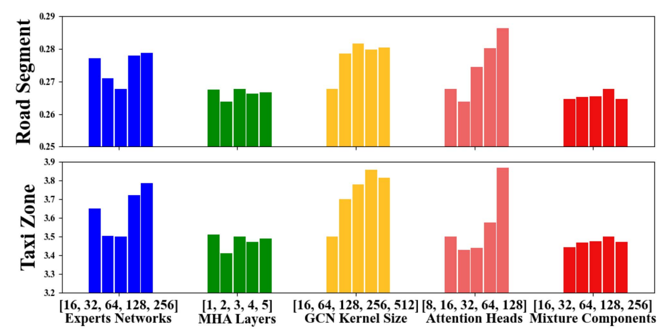


Fig. 8. RMSE under different hyperparameter settings.

the exact accident risk values, as they can support proactive traffic management and resource allocation. The success of our approach stems from the novel application of a zero-inflation mixture Poisson distribution, which effectively addresses the zero-inflation issue.

F. Hyperparameter Study

To evaluate the prediction performance of our model with different hyperparameter configurations, we select several key hyperparameters for experimentation. Specifically, we focus on the number of expert networks in SEM, the layers of MHA, the GCN kernel size, the number of attention heads, and the number of mixture components in ZIMP. All experiments are conducted on the Brooklyn dataset. Fig. 8 depicts the experimental results about RMSE for both granularities, while Fig. 9 depicts the experimental results for Recall and MAP across both granularities.

TABLE VII

COMPUTATION COST (WITH / TO SPLIT THE COST FOR COARSE-GRANULARITY AND FINE-GRANULARITY PREDICTION.)

Models	Training time (s/epoch)	Total training time (s)	Total GPU cost (MB)	Total inference time (s)	Total running time (s)
MLP	12/13	599/634	228/280	1/3	724/904
GRU	32/34	546/472	632/806	2/5	604/561
SDCAE	27/31	592/367	268/304	2/5	655/449
ConvLSTM	60/69	1203/1515	220/354	4/8	1334/1758
AGCRN	108/477	1829/6204	604/4898	11/65	2079/7218
GSNet	57/132	969/1582	376/2272	4/17	1054/1834
MVMT-STN	257	4115	2126	30	4747
MG-TAR	210/252	5877/4538	508/2450	18/27	6522/5158
MGHSTN	98/360	2264/8632	500/3648	8/33	2497/9629
STKGRisk	874	9618	5846	98	10921

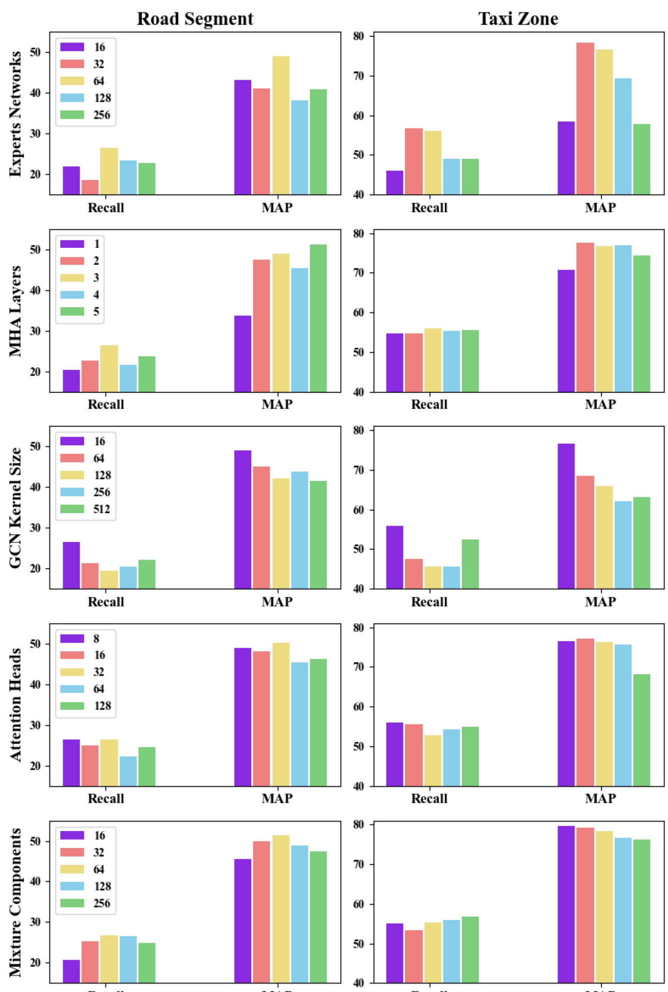


Fig. 9. Recall and MAP under different hyperparameter settings.

From Figs. 8 and 9, we observe that the number of MHA layers has the least impact on prediction performance. In contrast, both the GCN kernel size and the number of attention heads perform better with smaller values. Furthermore, the number of expert networks should neither be too small nor too large; medium values yield the best performance. The number of mixture components m has a small effect on RMSE but a large effect on Recall and MAP.

TABLE VIII

MODEL PERFORMANCE WITH AND WITHOUT ACCIDENT RISK SPATIAL-TEMPORAL PROPAGATION STRATEGY

Granularities	Models	RMSE↓	Recall↑	MAP↑	RMSE*↓	Recall*↑	MAP*↑
Road Segment	STKGRisk	0.2677	26.48	48.94	0.3132	29.81	56.90
	w/o Propagate	0.3986	0.00	0.00	0.4168	0.00	0.00
Taxi Zone	STKGRisk	3.5023	55.95	76.61	4.1378	59.95	81.47
	w/o Propagate	0.8142	17.26	26.69	0.9431	13.71	22.64

G. *Effects of Accident Risk Spatial-Temporal Propagation Strategy*

To evaluate the reasonableness and effectiveness of the spatiotemporal accident risk propagation strategy, we conduct corresponding ablation experiments on the Brooklyn dataset. The results are summarized in Table VIII. For fine-granularity prediction, all performance metrics deteriorate significantly when the propagation strategy is removed. This is because the original accident risk data, without propagation, is extremely sparse—causing the model to predict very low or even zero risk values. In the case of coarse granularity, removing the propagation strategy results in improved RMSE and RMSE* values, while other metrics decrease noticeably. The changes of RMSE and RMSE* are opposite to those in fine granularity is because each taxi zone typically aggregates data from multiple road segments, making the sparsity issue less severe. Since the risk value of each taxi zone is the sum of its constituent road segments, the propagation strategy increases the overall risk magnitude, thereby improving RMSE and RMSE*. However, it is important to note that in practical applications, identifying the most dangerous regions is often more critical than estimating exact risk values. The ablation of the propagation strategy results in a significant decline in both Recall and MAP, underscoring its critical role in enhancing model performance and practical applicability.

H. *Computation Cost*

We evaluate the efficiency of baseline models under the same setting, using an NVIDIA RTX A4000 GPU with a batch size of 2. As shown in Table VII, five types of cost are recorded: per-epoch training time, total training time, total runtime, total inference time, and GPU memory usage. Notably, MVMT-STN and STKGRisk are the only models that can simultaneously predict accident risk at both fine and coarse granularities, whereas the other models address the two tasks separately. Thus, their reported costs are divided into fine- and coarse-granularity prediction, and their total costs are the sum of both. Overall, fine-granularity prediction is more time- and memory-consuming than coarse-granularity prediction.

In addition, we provide a theoretical computational complexity analysis of STKGRisk. The overall complexity consists of three main parts: the STKG embedding module with complexity $\mathcal{O}(|\mathcal{F}|d)$, which can be pre-trained offline; the spatial-temporal graph modeling module with complexity $\mathcal{O}(N^2\ d + ED)$, where N is the number of regions, E is the number of edges, D is the kernel size of GCN, and in our paper $E \gg N$. The $\mathcal{O}(N^2\ d)$ term arises from deep-level graph construction and attention operations, and $\mathcal{O}(ED)$ corresponds to graph convolutions; the temporal modeling module with complexity $\mathcal{O}(TNd_h^2)$. Overall, the computational complexity is $\mathcal{O}(|\mathcal{F}|d + N^2\ d + ED + TNd_h^2)$. Finally, empirical results shown in Table VII, demonstrate that STKGRisk achieves accurate multi-granularity prediction within a unified framework while maintaining competitive efficiency in both computation time and GPU usage.

VI. Conclusion

In this paper, we propose STKGRisk, a multi-granularity, multi-level, and multi-view traffic accident risk prediction model. Specifically, we construct the first spatial-temporal knowledge graph (STKG) for traffic accidents and design a diachronic embedding module (DETA) to capture dynamic and high-order interactions of entities. To jointly model spatiotemporal correlations at both fine and coarse granularities, we introduce a multi-level, multi-view STGN encoder along with a message-sharing mechanism. To effectively address the zero-inflation issue in accident risk data, we adopt a zero-inflation Mixture Poisson (ZIMP) distribution to model its underlying distribution, also for the first time in this context. We conduct extensive experiments on three real-world multi-granularity traffic accident risk datasets. The results demonstrate the effectiveness and superiority of STKGRisk across multiple evaluation metrics and settings.

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